

# Fully Connected Neural Network Approach

## Assignment 5.3 - Deliverable Report

### Submitted By:

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## 1. Objective

The goal of this assignment was to implement a **Fully Connected Neural Network (FCNN/MLP)** to classify activities from the UCI Human Activity Recognition (HAR) dataset using extracted statistical, temporal, and frequency-domain features.

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## 2. Preprocessing & Setup

- **Dataset:** UCI HAR Dataset (7352 samples).
  - **Features extracted:** 16 features per sample, including statistical (mean, median, std, variance, min, max, skew, kurtosis), temporal (zero-crossing rate, energy, peak-to-peak, SMA), and frequency-domain (spectral centroid, spectral entropy).
  - **Labels:** 6 activity classes (0–5).
  - **Scaling:** StandardScaler applied to features.
  - **Train/Validation Split:** 80/20 stratified split.
  - **Data saved:** Features saved as `har_features.csv` for reuse.
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## 3. FCNN Architecture Design

- Input layer: 16 features
- Hidden layers:
  - 128 → ReLU
  - 64 → ReLU
  - 32 → ReLU
- Output layer: 6 neurons (logits)
- **Loss:** Cross-Entropy Loss
- **Optimizer:** Adam (learning rate 0.001)
- **Batch size:** 32, Epochs: 50

The architecture was chosen to balance complexity and generalization for the given feature set.

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## 4. Model Training & Results

- Training and validation loss decreased steadily, indicating proper learning.
  - Final **accuracy**: 68.4%
  - **F1 score**: 0.673
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## 5. Analysis & Discussion

### Training Curve Interpretation:

- Both training and validation loss decrease smoothly.
- No significant gap between training and validation loss, indicating the model is **neither overfitting nor underfitting** severely.

### Comparison with Classical ML (Assignment 5.2):

- Classical ML models achieved ~57% accuracy (best: Random Forest/XGBoost).
- FCNN improved accuracy to ~68%, showing that neural networks can better capture complex patterns in the features.

### Overfitting / Underfitting:

- No major overfitting: validation loss decreases with training loss.
- No major underfitting: training loss is reasonably low.
- Slight underfitting may exist for some classes with overlapping patterns, which could be improved with deeper networks, dropout, or additional features.

### Limitations:

- FCNN relies on pre-extracted features, may miss temporal patterns in raw sequences.
  - Accuracy could further improve with feature engineering, more layers, or sequence models like LSTM.
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## 6. Summary

- Successfully implemented a **FCNN for activity classification** using HAR dataset.
- Achieved **68% accuracy** and **F1 score of 0.673**, outperforming classical ML models.

- Model is **saved** (`fcnn_model.pth`) and features (`har_features.csv`) are reusable.
- Architecture demonstrates proper learning with minimal overfitting and provides a basis for extending to more advanced deep learning models.